

Stage 2 Report

Flow Matching to Class Prototypes

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We insert a flow-matching layer between the frozen image feature and the prototype classifier built in Stage 1. A velocity network transports each feature toward the prototype of its class; the transported feature is then classified with the same cosine rule as before. Everything downstream of the encoder is reused unchanged, so any accuracy difference is attributable to the FM layer alone.

This report follows the same order as the work itself: a Stage 1 refresher, the brief's requirements, then one section per pipeline step.

The three headline results

- **The FM layer works** — up to **+22.3 pp** over the prototype baseline (DINOv2/Aircraft, full data). 24 of 36 cells improve.
- **Rolled-out training does not beat standard FM** — it wins only 4 of 18 head-to-head cells, despite removing the train/inference mismatch it was designed to remove.
- **One quantity predicts every cell** — the gain tracks the headroom Stage 1 left behind (Spearman $\rho = 1.000$), and explains the two cells where FM loses.

Where Stage 1 left off

Two pages of context: the protocol Stage 2 inherits, and the numbers it has to beat.

Stage 1 froze two pretrained encoders, cached their features once, and built two classifiers on top without ever fine-tuning the encoder. Stage 2 changes exactly one thing about that setup — it inserts a learned transport before the classification rule. The encoder, the cached features, the k-shot subsets, the seeds, the test splits and the prototypes are all reused unchanged, so every number in this report reads as a difference against the prototype column below with nothing else varying.

The protocol, unchanged in Stage 2

What was used	
Datasets	DTD (47 texture classes, partition 1) · FGVC-Aircraft (100 variants)
Encoders	ResNet-18 (512-d, both datasets) · DINOv2 ViT-S/14 (384-d, Aircraft)
Combinations	3 encoder/dataset pairs carried through every experiment
Training sizes	K = 5, 10, full · 3 balanced random subsets per K, seeds 0/1/2
Test set	the complete official test split (1880 / 3333 / 3333), never subsampled
Reported as	top-1 accuracy, mean $\pm$ sample std over 3 seeds

The two Stage 1 baselines

Linear probe — a trained classifier on the frozen features, $s = Wz + b$. It is the reference for how much the features can give when a classifier is actually fitted.

Prototype classifier — one mean feature vector per class, no training at all, classify by cosine similarity to the nearest prototype. **This is Stage 2's baseline**, and its prototypes are the targets the flow transports toward.

The numbers Stage 2 has to beat

Encoder / dataset	K	Linear probe	Prototype (Stage 2 baseline)	Headroom
ResNet-18 / DTD	5	44.86 $\pm$ 1.83	46.10 $\pm$ 1.89	-1.24
ResNet-18 / DTD	10	52.04 $\pm$ 1.16	51.67 $\pm$ 1.33	+0.37
ResNet-18 / DTD	full	62.82 $\pm$ 0.42	58.83 (1 run)	+3.99
ResNet-18 / Aircraft	5	20.76 $\pm$ 0.26	16.46 $\pm$ 0.39	+4.30
ResNet-18 / Aircraft	10	27.87 $\pm$ 1.33	20.19 $\pm$ 1.14	+7.68
ResNet-18 / Aircraft	full	38.15 $\pm$ 0.29	25.47 (1 run)	+12.68
DINOv2 / Aircraft	5	38.19 $\pm$ 2.49	23.79 $\pm$ 0.47	+14.40
DINOv2 / Aircraft	10	51.54 $\pm$ 1.70	27.74 $\pm$ 0.98	+23.80
DINOv2 / Aircraft	full	67.55 $\pm$ 0.28	34.26 (1 run)	+33.29

Headroom is the last column: how much a *trained* classifier gains over the untrained prototype on the same cell. It is small or negative on DTD (the class mean is already a good summary of a texture) and very large on DINOv2/Aircraft. Keep this column in mind — it turns out to predict the Stage 2 result exactly, and it is the last thing in this report.

What Stage 2 asked for

Every requirement in the assignment, and where in this report it is answered.

The two training objectives

	Standard FM	Rolled-out FM
Sampling	$t \sim U(0,1)$, $z_t = (1-t)z_i + t \cdot p$	full T-step Euler rollout from z_i
Target	velocity $u_i = p - z_i$	final position p
Loss	$\ v(z_t, t) - u_i\ ^2$	$\ \hat{z}_T - p\ ^2$
Backprop through	one velocity prediction	all T predictions
Train/inference	mismatched	matched
Models needed	1 (serves every T)	2 (one per T)

Inference is identical for both: T Euler steps $\hat{z}(k+1) = \hat{z}k + (1/T) \cdot v(\hat{z}k, k/T)$ for $k = 0 \dots T-1$, then classify $\hat{z}_T$ by cosine similarity to the same prototypes. Evaluated at **T = 4 and T = 12**.

Required deliverables

#	What the brief asks for	Where
1	Accuracy table, ΔAcc vs. baseline, accuracy-vs-K plot with error bars	Step 04
2	Representative training-loss curves, both objectives	Step 04
3	Feature space: original vs. standard FM vs. rolled-out, jointly-fitted projection	Step 05
4	Flow trajectories: intermediate states, original, transported, prototype	Step 05
opt.	The learned flow in reverse, starting from prototypes	Step 05
opt.	Samples vs. prototypes at intermediate flow times	Step 05

Protocol constraints, and how they were met

- *Same datasets, encoders, prototypes, subsets, seeds, test sets as Stage 1* — step 01 re-derives the Stage 1 baseline from scratch and reproduces all 21 runs **exactly** (max $|\Delta| = 0.00e+00$); the prototypes are byte-identical to Stage 1's saved file.
- *Keep architecture and main training choices fixed between the two objectives* — both go through one shared setup function, so for a given seed they start from identical weights and see identical batch order. Gradient clipping was applied to **both**, not only the one that needed it.
- *Same repetition protocol as Stage 1* — 3 seeds everywhere. The Stage 1 full-data baseline is a single deterministic run (nothing left to subsample), so it carries no error bar; every other cell does.
- *No extensive hyperparameter optimisation* — the MLP is the brief's suggestion (2×512 , SiLU, t concatenated) and was not tuned.

The experimental grid

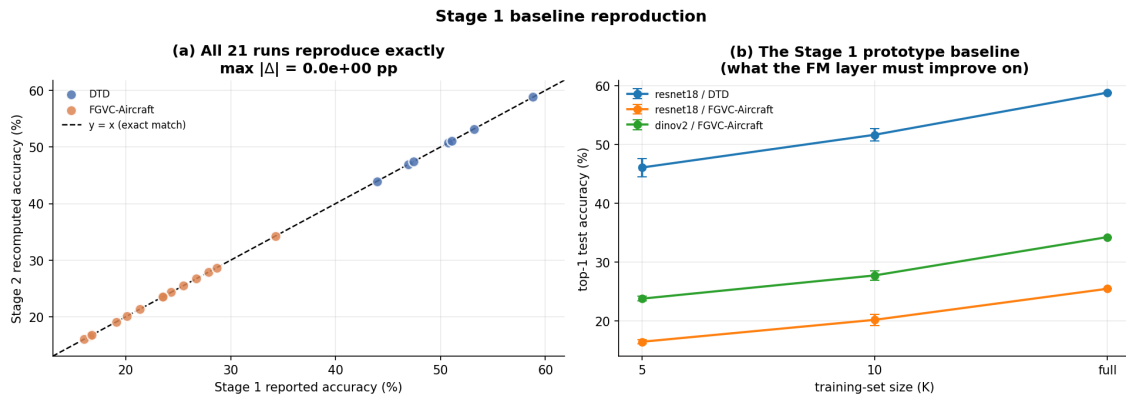
3 encoder/dataset combinations $\times$ 3 training sizes $\times$ 3 seeds. Standard FM needs one model per cell (it is T-independent); rolled-out needs one per T. That is **81 trained models** filling a **45-cell** results table (3 combos $\times$ 3 K $\times$ 5 methods).

01 · Setup and prototypes

Rebuild Stage 1's baseline from scratch, verify it, and settle the one design decision Stage 2 has to make before writing any training code.

Before building on Stage 1, reproduce it

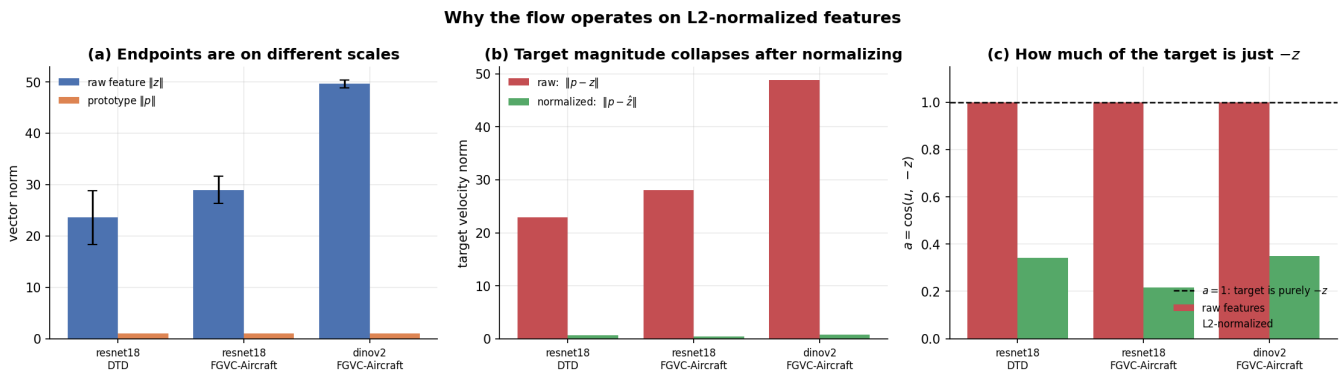
The whole comparison rests on the Stage 2 baseline column being the *same* quantity Stage 1 reported. So step 01 recomputes all 21 prototype runs from the cached features rather than copying the numbers across.



All 21 runs land on the $y = x$ line: $\max |\Delta| = 0.00e+00$. The prototypes themselves are byte-identical to Stage 1's saved file, so the flow is aimed at exactly the targets Stage 1 built.

The one decision: L2-normalize before the flow

Prototypes are unit-norm by construction. Raw encoder features are not — they sit at norms of 24-50. The flow target is $u = p - z$, so when $\|z\| \approx 30$ and $\|p\| = 1$, the prototype is lost in the rounding and u is very nearly just $-z$.

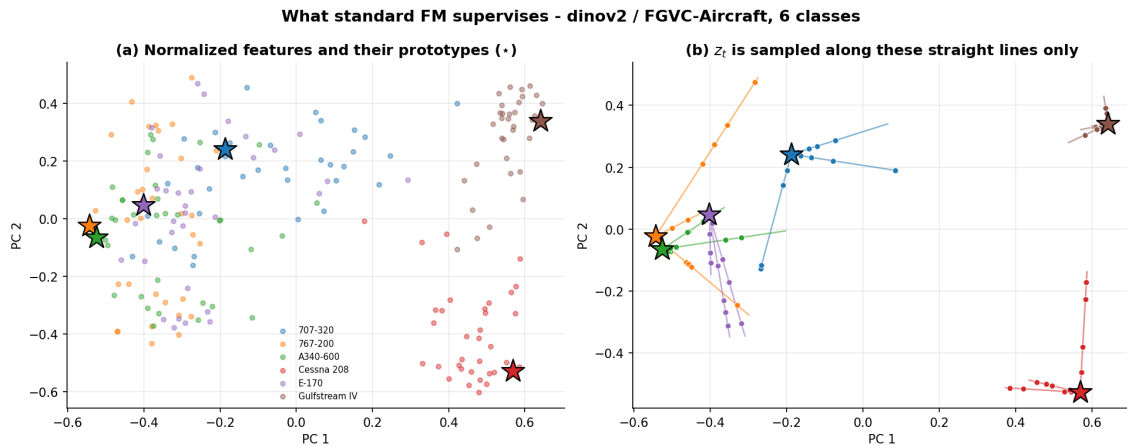


Measured with $a = \cos(u, -z)$, the alignment between the supervised direction and 'just cancel the feature'. On raw features $a = 0.9995-0.9999$ — the target carries almost no class information. After normalizing it drops to 0.22-0.35.

This is not left as an argument. Step 02 retrains the identical objective with normalization switched off. Raw features cost **15.8-20.3 pp**, and on both ResNet-18 combinations the flow ends up *worse than no flow at all*. The prediction from geometry and the measured outcome agree.

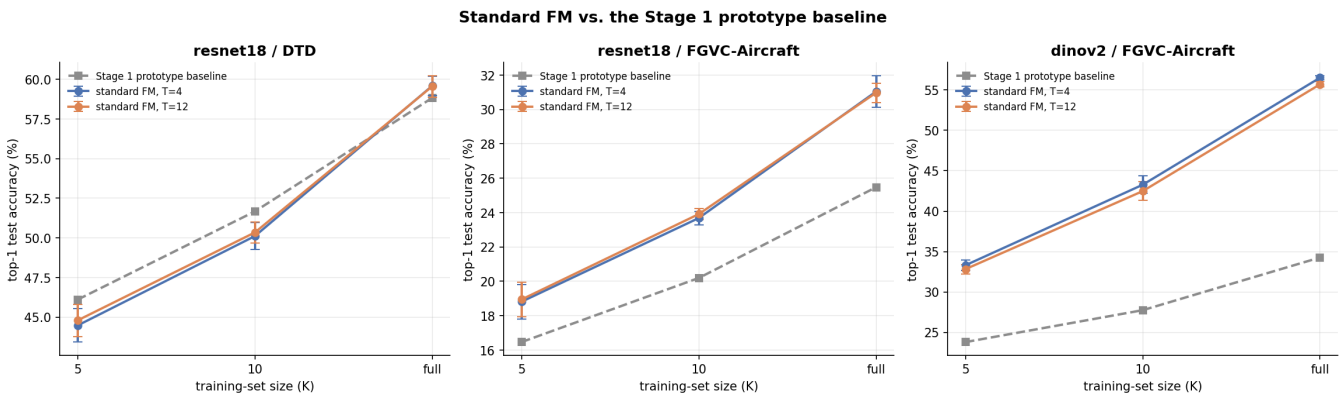
02 · Standard flow matching

Supervise the velocity at random points on the ideal straight path between a feature and its prototype.



The objective made visible. Training samples a random t , places z_t on the straight line from feature to prototype, and asks the network for that line's constant velocity $u = p - z$.

What we get

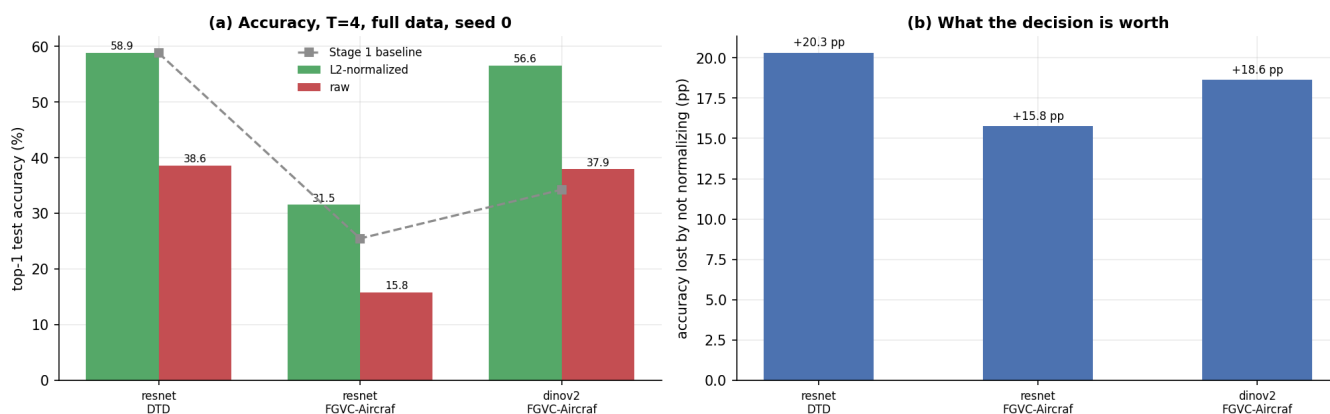


Standard FM at both T against the Stage 1 baseline, with error bars over 3 seeds. **Large gains on Aircraft, essentially none on DTD** — and at $K = 5$ and 10 on DTD the layer is slightly harmful.

- Best cell: **DINOv2/Aircraft, full data, +22.3 pp** (34.26 → 56.52).
- Worst cell: ResNet-18/DTD at $K = 10$, **−1.6 pp**.
- $T = 4$ and $T = 12$ differ by under 1 pp everywhere, with inconsistent sign.

Was the normalization decision actually necessary?

Ablation: L2 normalization is load-bearing, not cosmetic

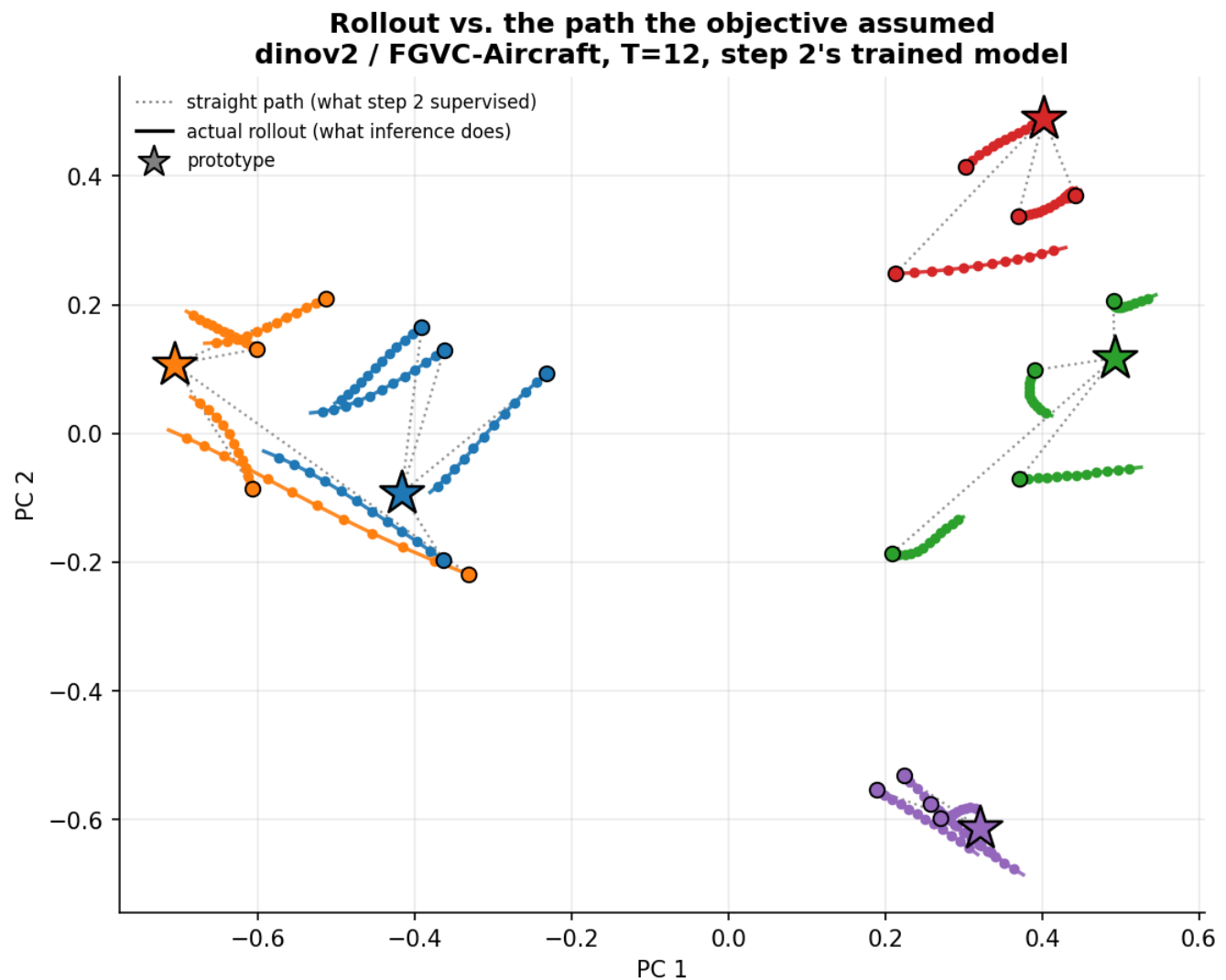


The step 01 decision, measured rather than argued. Training the same objective on raw features costs 15.8–20.3 pp. The loss tells the story: it settles at **0.08–0.20** on unit-norm inputs and at **27–111** on raw ones.

03 · Rolled-out flow matching

Train on exactly the states inference produces, and supervise only where the feature ends up.

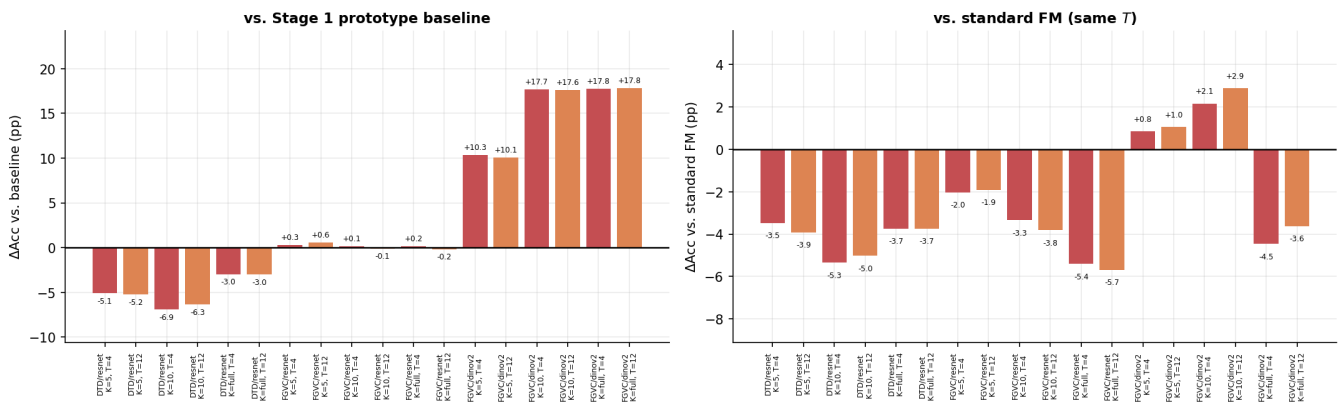
The problem it is designed to fix



Standard FM only ever sees points on the ideal straight line. At inference the network is fed states generated by its own previous predictions — and those drift off that line. Nothing in its training taught it how to recover.

What we get — it does not help

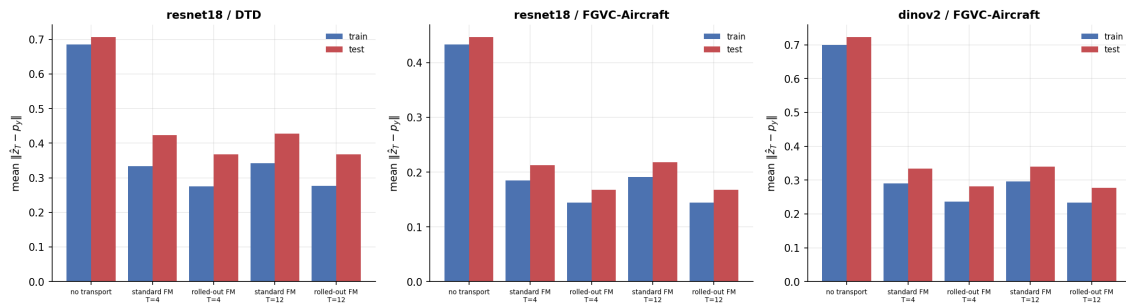
Rolled-out FM - change relative to each reference



Rolled-out FM against the baseline (left) and against standard FM (right). It wins only **4 of 18** head-to-head cells, and loses badly on DTD (down to -6.9 pp).

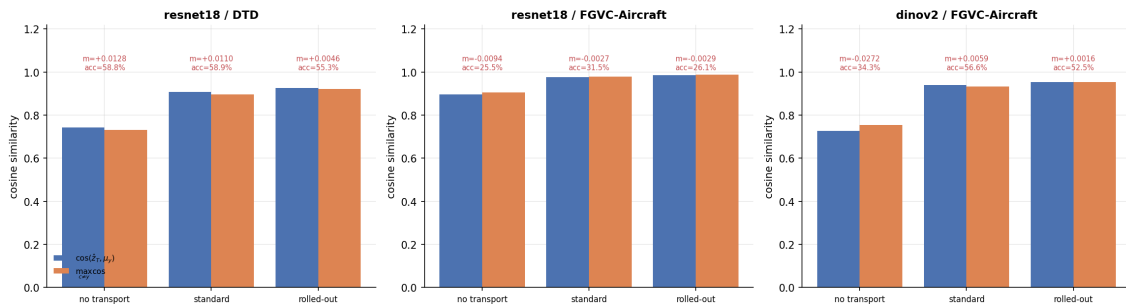
Why not? It transports better but decides worse

How close the transport gets to the prototype - train vs. test (full data, seed 0)



Distance to the target prototype, train **and** test. Rolled-out FM reaches the prototype more closely on test data too — so this is not overfitting. It is doing its job well.

Rolled-out FM raises similarity to the correct prototype AND to its competitors (T=4, full data, seed 0)



The catch: it raises similarity to the correct prototype **and** to the best competing one by nearly the same amount. A better transport is not a better decision — only the gap between the two decides the label.

One run diverged. Backpropagating through T composed Euler steps can amplify gradients; 1 of 54 rolled-out runs blew up late in training, while standard FM never did in 27. We added gradient-norm clipping at 1.0 — and applied it to **both** objectives, not just the one that needed it, so the two stay configuration-identical. It is inert for standard FM (every cell moved by ≤ 0.25 pp). The unclipped run is kept as evidence.

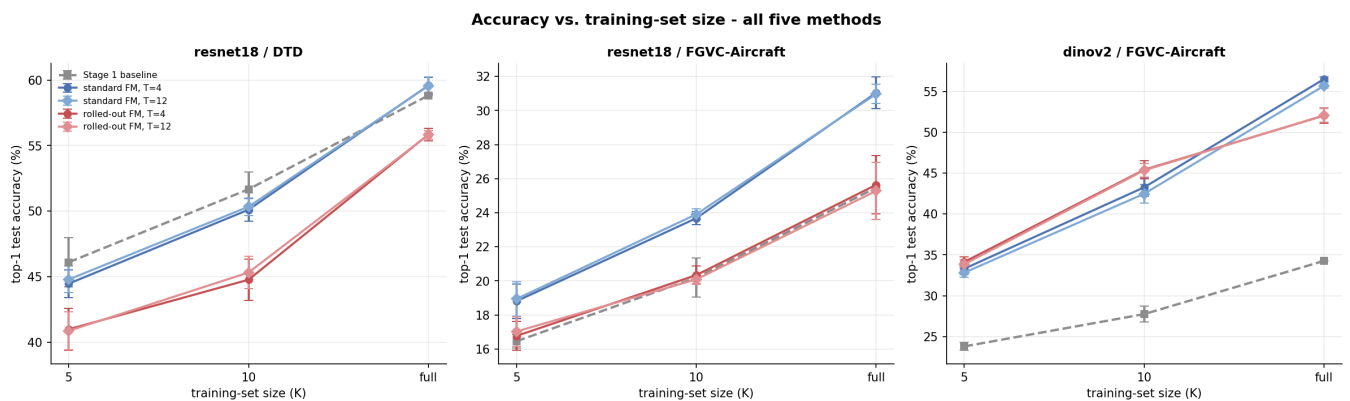
04 · Evaluation

Deliverables 1 and 2: the complete accuracy table, ΔAcc , accuracy vs. K with error bars, and training curves.

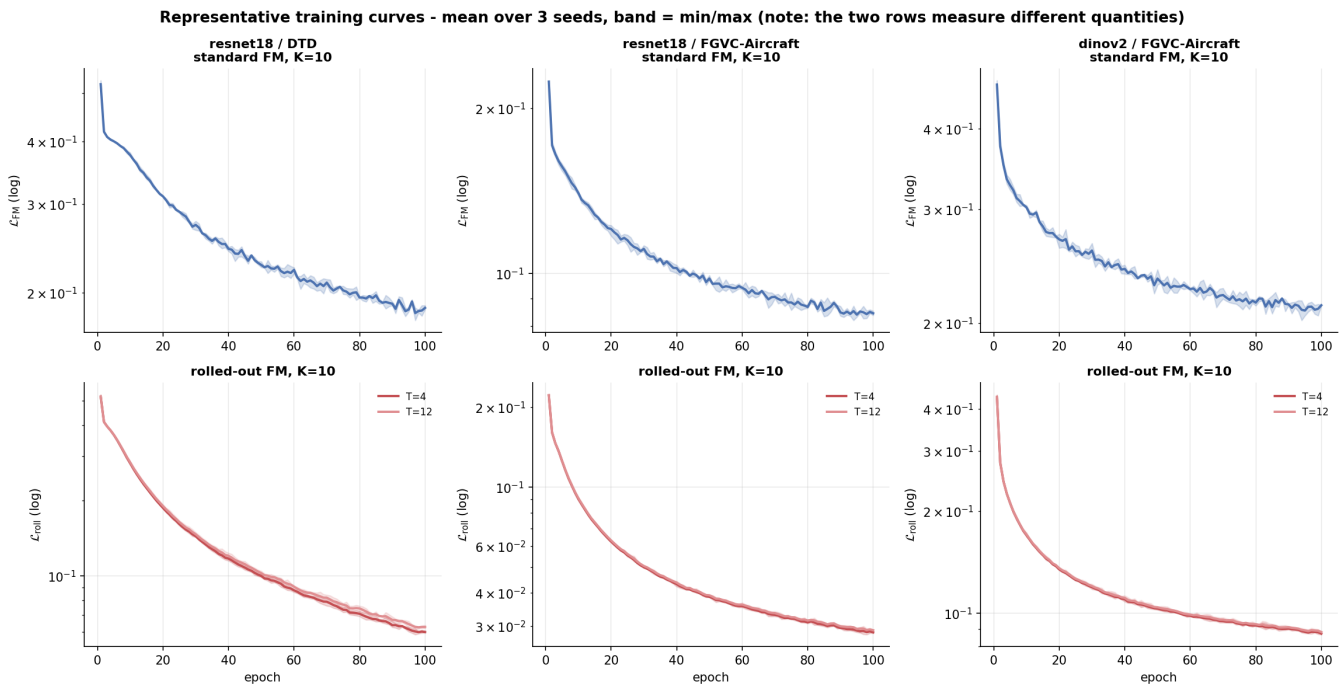
The complete 45-cell table (deliverable 1)

Encoder / dataset	K	Baseline	std T=4	std T=12	roll T=4	roll T=12
ResNet-18 / DTD	5	46.10 $\pm$ 1.89	44.47 $\pm$ 1.05	44.79 $\pm$ 1.01	40.99 $\pm$ 1.60	40.87 $\pm$ 1.46
ResNet-18 / DTD	10	51.67 $\pm$ 1.33	50.11 $\pm$ 0.86	50.34 $\pm$ 0.66	44.77 $\pm$ 1.58	45.32 $\pm$ 1.23
ResNet-18 / DTD	full	58.83 (1 run)	59.59 $\pm$ 0.61	59.56 $\pm$ 0.68	55.85 $\pm$ 0.47	55.82 $\pm$ 0.32
ResNet-18 / Aircraft	5	16.46 $\pm$ 0.39	18.81 $\pm$ 1.01	18.94 $\pm$ 1.02	16.77 $\pm$ 0.84	17.03 $\pm$ 0.87
ResNet-18 / Aircraft	10	20.19 $\pm$ 1.14	23.68 $\pm$ 0.39	23.91 $\pm$ 0.32	20.34 $\pm$ 0.53	20.09 $\pm$ 0.24
ResNet-18 / Aircraft	full	25.47 (1 run)	31.04 $\pm$ 0.93	30.97 $\pm$ 0.56	25.64 $\pm$ 1.72	25.28 $\pm$ 1.69
DINOv2 / Aircraft	5	23.79 $\pm$ 0.47	33.30 $\pm$ 0.68	32.82 $\pm$ 0.57	34.13 $\pm$ 0.59	33.86 $\pm$ 0.58
DINOv2 / Aircraft	10	27.74 $\pm$ 0.98	43.27 $\pm$ 1.09	42.48 $\pm$ 1.16	45.42 $\pm$ 1.10	45.35 $\pm$ 0.83
DINOv2 / Aircraft	full	34.26 (1 run)	56.52 $\pm$ 0.27	55.70 $\pm$ 0.31	52.05 $\pm$ 0.91	52.08 $\pm$ 0.82

Bold marks the best FM cell in a row **when it beats the baseline** — three rows have no bold because no flow configuration does. **24 of 36 FM cells improve**, from +22.26 pp down to −6.90 pp.



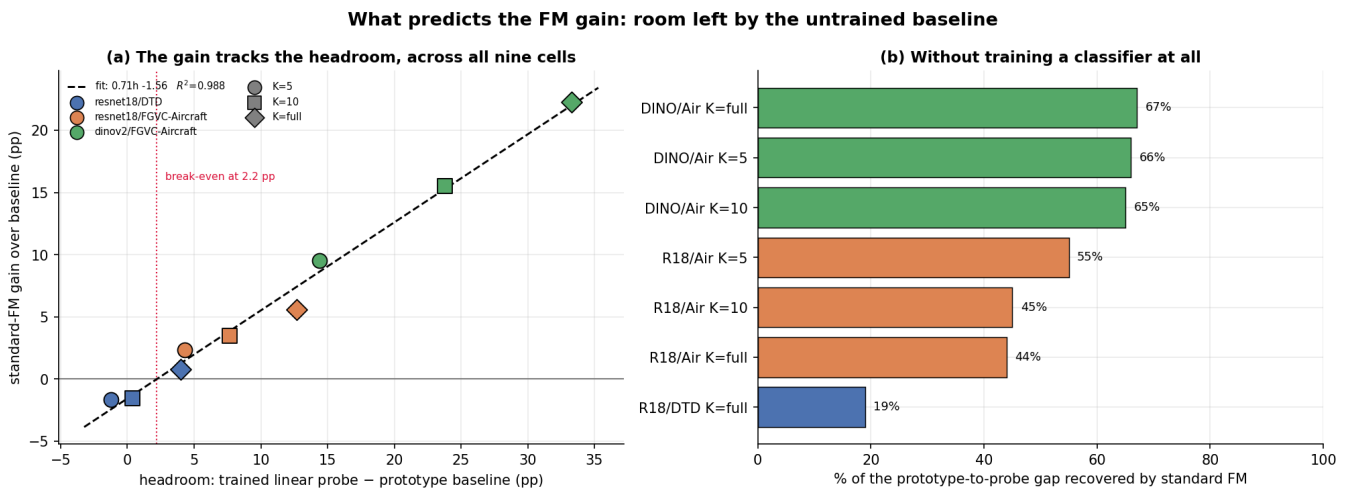
Deliverable 1. All five methods against training-set size, error bars over 3 seeds. The three panels are three different stories: FM is flat-to-harmful on DTD, clearly positive on ResNet-18/Aircraft, and transformative on DINOv2.



Deliverable 2. Loss per epoch for both objectives, mean over 3 seeds with a min/max band. Both converge smoothly and neither is still improving at the budget's end. The two are on different scales because they measure different things — velocity error vs. endpoint distance.

So when does the FM layer actually help?

The gains range from -6.9 to $+22.3$ pp. Feature separability does not order them, and K orders them only *within* a combination. The quantity that orders all nine cells at once is the **headroom** from the Stage 1 refresher — how much a trained linear probe gained over the untrained prototype on that same cell.



Spearman rho = 1.000 — the nine cells rank identically by headroom and by gain — with $R^2 = 0.988$. The fit breaks even at 2.2 pp of headroom, and the only two cells below that line are exactly the two where FM loses.

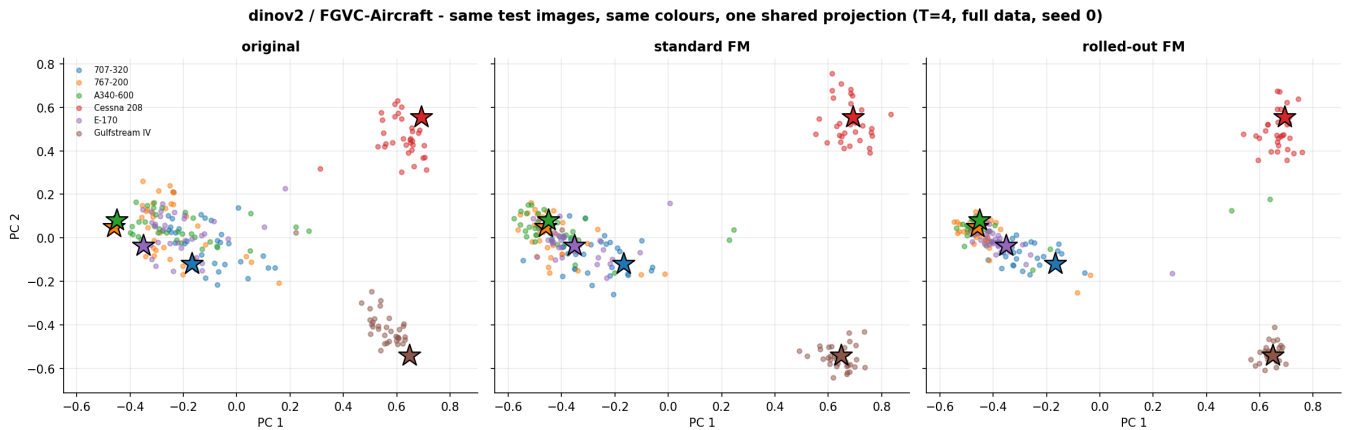
Read plainly: the FM layer recovers roughly **70 % of whatever a trained linear probe would have gained** over the prototype baseline, minus a fixed cost of about 1.6 pp — while keeping the prototype structure and training no classifier at all. It also explains the DTD failures: there the prototype baseline is already level with the trained probe, so there is nothing left for a transport to recover. Two caveats we would state out loud: nine cells is a small sample, and the linear probe is a *reference point*, not a mechanism.

05 · Feature-space visualizations

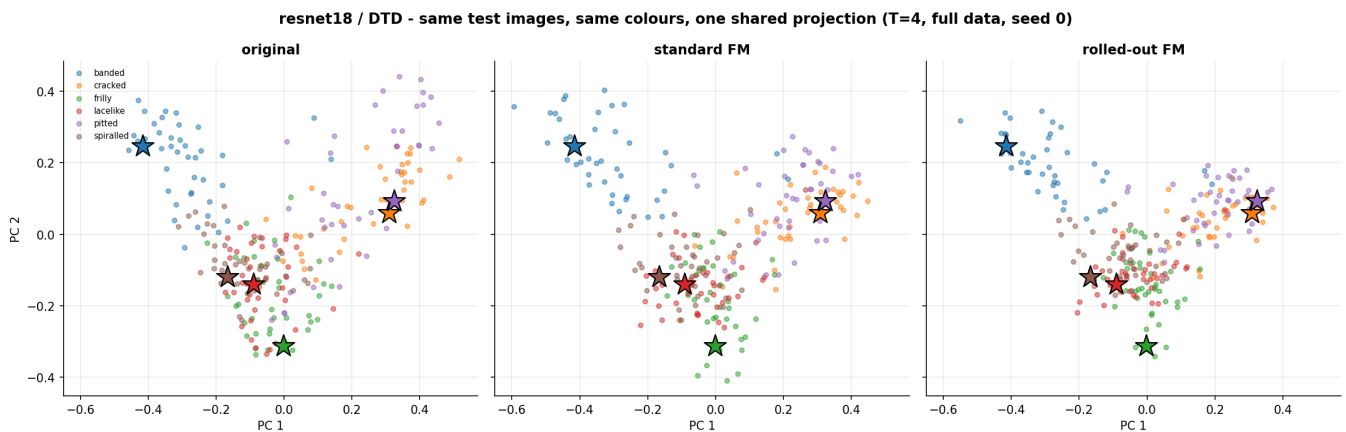
Deliverables 3 and 4, plus both optional explorations.

Feature space before and after (deliverable 3)

Same test images, same class colours, and the prototypes drawn in every panel. One PCA is fitted on the **concatenation** of all three feature sets and the prototypes, so the panels are directly comparable rather than three independent rotations.

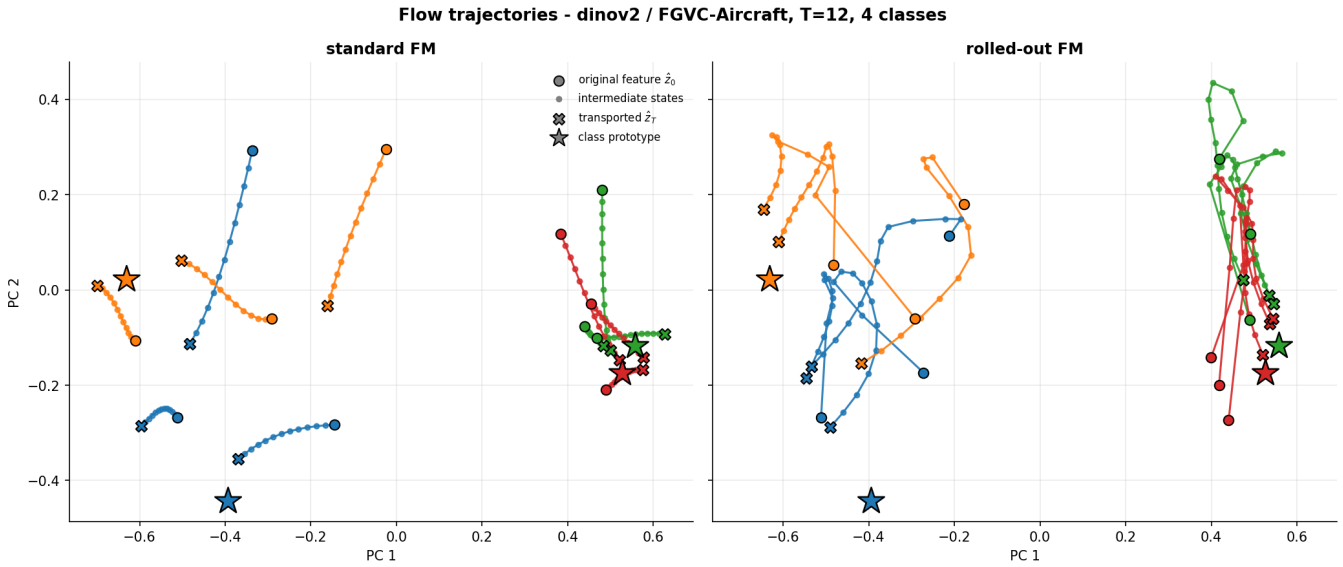


The success case (+22.3 pp). Both flows contract each class toward its prototype (stars), rolled-out most of all. But look at the four overlapping classes on the left: they stay overlapping in all three panels.



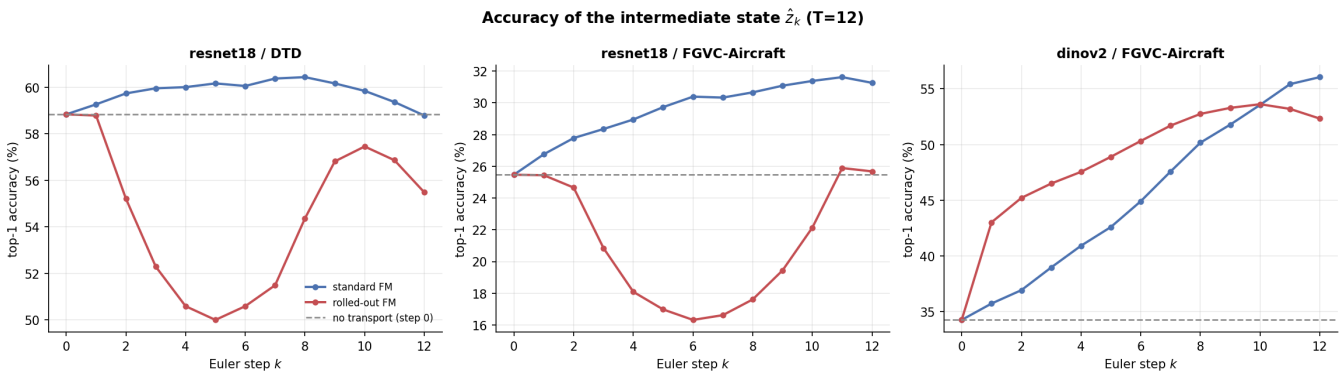
The failure case (−1.6 pp) — and it looks much the same. **Contraction toward the right prototype is not the same thing as separation from the wrong ones**, which is why the pictures alone cannot tell you which case you are looking at.

Flow trajectories (deliverable 4)



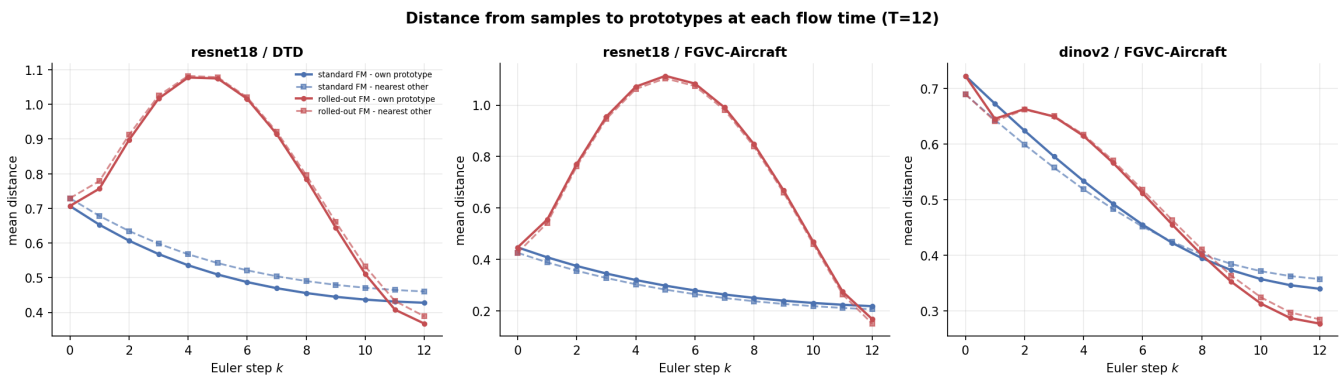
Original feature (circle), intermediate states (small dots), transported feature (cross), class prototype (star), under a joint PCA. **Standard FM takes smooth, near-straight paths; rolled-out FM zigzags and loops.** Only $\hat{z}_T$ is supervised, so nothing constrains the path it takes to get there.

Accuracy along the rollout



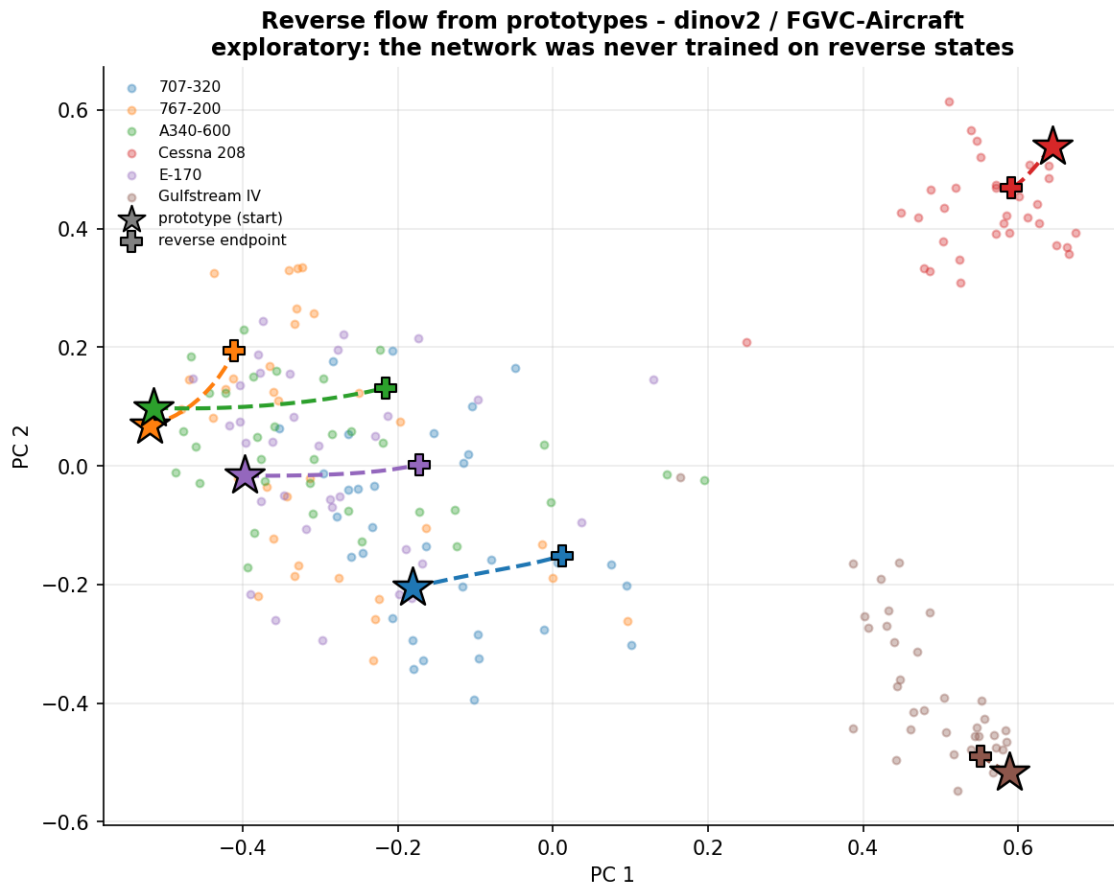
Classifying the intermediate state at every Euler step. Rolled-out accuracy collapses mid-rollout — on ResNet-18/Aircraft the exact midpoint is **9.2 pp worse than doing nothing at all** — then recovers by the final step.

Samples vs. prototypes at intermediate flow times (optional)



Distance to the **own** prototype (solid) and to the **nearest competing** one (dashed), per step. Three things at once: rolled-out FM moves features up to **149 % further away** before returning; it ends closer to the target than standard FM in all three; and the solid and dashed curves stay glued together, so only the gap survives — and standard FM's gap is wider everywhere.

The flow in reverse (optional)



Running the learned field backwards from the prototypes. The flow is a learned transport, not an invertible generative map — reversing it does not recover the feature distribution it came from.

What we conclude

And what we checked before concluding it.

Answers to the brief's three questions

Question	Answer
Does FM beat the prototype baseline?	Yes, where there was room — up to +22.3 pp. 24 of 36 cells improve.
Does rolled-out beat standard FM?	No. It wins 4 of 18 cells despite removing the mismatch it targets.
How does it change the geometry?	It contracts each class toward its prototype — but toward rivals too.

Five findings

- **The FM layer works, and works best where prototypes were weakest.** Up to +22.3 pp.
- **Removing the train/inference mismatch does not help.** Rolled-out FM reaches the prototype more closely — on test data, so not overfitting — while raising similarity to competing prototypes just as much. A better transport, not a better decision.
- **The number of Euler steps barely matters.** Under 1 pp between $T = 4$ and $T = 12$ everywhere. For rolled-out FM, where T is part of training, $3\times$ the cost buys nothing.
- **Rolled-out flows take meaningless detours** — up to 149 % further from the target mid-rollout, and 9.2 pp less accurate there than doing nothing.
- **One quantity predicts every cell.** Gain tracks the headroom Stage 1 left behind ($\rho = 1.000$, $R^2 = 0.988$); the FM layer recovers ~ 70 % of it, and the two cells below the 2.2 pp break-even are exactly the two where FM loses.

What we verified before trusting any of it

- **The code matches the brief.** A test suite re-derives each of the four formulas from the brief's wording and checks the implementation against it — 21 tests, all passing. We validated the suite by breaking the code on purpose and confirming it noticed.
- **Stage 1 reproduces exactly.** All 21 baseline runs, $\max |\Delta| = 0.00e+00$; prototypes byte-identical.
- **Every result is deterministic.** Retraining all 81 models from scratch reproduced every accuracy and every loss curve bit-for-bit.
- **Reported accuracies match the saved predictions.** All 108 runs recomputed from stored per-example predictions, max difference $6e-06$ pp.
- **The evaluation table cannot silently disagree with its own inputs.** The notebook asserts all 36 FM cells match steps 02–03 and all 9 baseline cells match Stage 1's published table, and fails rather than publishing a table that drifts.

Questions we expect, and where the answer is

If asked...	Look at
Why cosine and not Euclidean nearest-prototype?	With unit-norm prototypes they are identical
Why does rolled-out FM lose if it removes the mismatch?	Step 03 margins; step 05 distance-by-step
Did you tune anything to get these numbers?	No — brief's architecture, fixed epochs, no selection
Why has the full-data baseline no error bar?	Nothing left to subsample; one deterministic run
Is the L2 normalization doing the work?	Step 02 ablation: raw costs 15.8–20.3 pp